FATIMA JINNAH WOMEN UNIVERSITY

Hospital Management System

(Report)

Submitted to: Sir Adeel Khalid

Submitted by:

1. Javeria Rahim(2023-BCS-046)
2. Romaisa(2023-BCS-073)
3. Malaika(2023-BCS-056)
4. Ayman Un Nisa(2023-BCS-023)
5. Maryam Zulfiqar(2023-BCS-063)
6. Farwa Batool(2023-BCS-029)

Department of Computer Science.

FATIMA JINNAH WOMEN UNIVERSITY

RAWALPINDI

www.fjwu.edu.pk

BACKGROUND:

HOSPITAL
MANAGEMENT SYSTEM

The Hospital Management System is designed to streamline and automate various tasks related to patient admission, discharge, and doctor information. This system facilitates efficient management of patient and doctor records, providing a centralized platform for hospital staff to perform essential operations. It enables doctors and admin to view and modify appointments schedules if required. The purpose of this project is to computerize all details regarding patient details and hospital details. The system will be used as the application that serves hospitals, clinics, dispensaries, or other health institutions. The intention of the system is to increase the number of patients that can be treated and managed properly. If the hospital management system is file based, the management of the hospital must put much effort into securing the files. They can be easily damaged by fire, insects, and natural disasters. Also, could be misplaced by losing data and information. The code is implemented in C, utilizing basic file handling and console-based user interaction.

INTRODUCTION:

The Hospital Management System is powerful, flexible, and easy to use and is designed and developed to deliver real conceivable benefits to hospitals. The Hospital Management System is designed for multispecialty hospitals, to cover a wide range of hospital administration and management processes. It is an integrated end-to-end Hospital Management System that provides relevant information across the hospital to support effective decision making for patient care, hospital administration and critical financial accounting, in a seamless flow. Hospital Management System is a software product suite designed to improve the quality and management of hospital management in the areas of clinical process analysis and activity-based costing. The Hospital Management System enables you to develop your organization and improve its effectiveness and quality of work. Managing the key processes efficiently is critical to the success of the hospital helps you manage your processes. Information about Patients is done by just writing the Patients name, age, and gender. Whenever the Patient comes up his information is stored freshly. Information about various diseases is not kept as any document. Doctors themselves do this job by remembering various medicines.

PROBLEM STATEMENT:

In this busy world we don't have the time to wait in infamously long hospital queues. The problem is, queuing at hospital is often managed manually by administrative staff, then take a token there and then wait for our turn then ask for the doctor and the most frustrating thing - we went there by traveling a long distance and then we come to know the doctor is on leave or the doctor can't take appointments. MS will help us overcome all these problems because now patients can book their appointments at home, they can check whether the doctor they want to meet is available or not. Doctors can also confirm or decline appointments, this help both patient and the doctor because if the doctor declines' appointment, then patient will know this in advance a patient will visit hospital only when the doctor confirms' the appointment this will save time and money of the patient. Patients can also pay the doctor's consultant fee online to save their time. HMS is essential for all healthcare establishments, be it hospitals, nursing homes, health clinics, rehabilitation centers, dispensaries, or clinics. The main goal is to computerize all the details regarding the patient and the hospital. The installation of this healthcare software results in improvement in administrative functions and hence better patient care, which is the prime focus of any healthcare unit.

METHODOLOGIES:

1. Admitting Patients:

The process of admitting patients involves capturing vital information such as ID, name, address, disease, and admission date. The admission date is automatically generated using the system date. The system utilizes C's file handling capabilities to store patient records persistently in a file named "patient.txt."

2. Displaying Patient List:

Patient information retrieval and display are achieved through reading data from the "patient.txt" file. The system presents patient records in a well-formatted manner, including details such as ID, name, address, disease, and admission date.

3. Discharging Patients:

Patient discharge is facilitated by entering the patient's ID. The system identifies the patient, removes their record from the "patient.txt" file, and ensures data integrity by using a temporary file during the process.

4. Adding Doctors:

Like patient admission, the system captures comprehensive details about doctors, including ID, name, specialization, address, and the date of addition. Doctor information is stored persistently in a file named "doctor.txt" using C's file handling functions.

5. Displaying Doctor List:

The system retrieves and displays doctor information from the "doctor.txt" file. The doctor list includes essential details such as ID, name, address, specialization, and the date of addition.

CONCLUSION:

In conclusion, the Hospital Management System, implemented in C, serves as a foundational step towards improving healthcare management processes. While the current version provides essential features for patient and doctor management, future iterations could explore additional functionalities such as search capabilities, data updates, and further error handling mechanisms. The system exemplifies a commitment to leveraging technology to enhance healthcare administration, setting the stage for more sophisticated solutions in the evolving landscape of hospital management systems.